

Adaptive Traffic Signal Control for Urban Intersections Using Riccati-Based Linear Quadratic Regulator Enhanced by LSTM-Based Traffic Flow Prediction

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Abstract—Efficient traffic signal control is essential for mitigating congestion and enhancing urban mobility. This study presents a hybrid control framework for coordinating signalized intersections, combining adaptive Linear Quadratic Regulator (LQR) with deep learning-based traffic state prediction. The traffic network is modeled as a discrete-time state-space system, where signal timings are optimized in real time via online solutions to the Riccati equation. A pre-trained Long Short-Term Memory (LSTM) network forecasts short-term traffic states, such as vehicle density and speed, which informs and improves the controller's responsiveness to upcoming conditions. Simulations conducted in the Simulation of Urban MOBility (SUMO) environment, based on a realistic sketch of a high-density urban roundabout in Ho Chi Minh City, demonstrate that the proposed method outperforms traditional Riccati and Riccati+LQR controllers. Compared to the Riccati baseline, the LSTM-enhanced controller achieves a 7.9% increase in average speed, a 29.39% reduction in average trip duration, a 57.97% decrease in waiting time, and a 42.64% reduction in time loss, while maintaining consistent vehicle throughput. These results highlight the potential of combining classical optimal control with deep learning for scalable, data-driven traffic management in smart cities.

Index Terms—Traffic Signal Control, LSTM Neural Networks, Optimal Control, Intelligent Transportation Systems, Microsimulation Analysis.

I. INTRODUCTION AND RELATED WORK

Urban traffic congestion leads to increased travel times, fuel consumption, and pollution. With rising populations and vehicle ownership, efficient traffic management is increasingly critical [1]. Traffic signal control remains a key strategy for optimizing intersection flow and mitigating congestion [2].

Traditional fixed-time and actuated signal plans often lack responsiveness to real-time traffic variations, limiting effectiveness under dynamic urban conditions [3]. Model-based methods such as the LQR, relying on state-space models and

the Riccati equation, provide systematic feedback control to minimize delays and maintain stability [4], [5], [6]. Despite their benefits, challenges remain in scalability and adaptability for large, complex networks [2].

Recent advances in deep learning, especially LSTM and Convolutional Neural Networks (CNNs), have significantly improved short-term traffic forecasting by capturing nonlinear spatiotemporal dependencies from sensor and crowd-sourced data [7], [8]. Integration of these predictive models with classical control frameworks enhances both adaptability and stability, enabling hybrid systems that combine foresight with feedback control [9], [10].

While actuated control adapts signals locally based on sensor inputs [11], it typically operates reactively without network-level coordination or predictive capabilities [12]. Deep learning models outperform traditional statistical approaches in traffic prediction and have been incorporated into control systems via Model Predictive Control (MPC) or reinforcement learning [13], [14], [15], [16].

Hybrid approaches leveraging LSTM-based forecasts to guide Riccati/LQR or MPC controllers provide proactive, stable traffic management [17], [18], though most focus on isolated intersections or small networks and lack continuous real-time adaptation of control parameters.

Addressing these gaps, this study proposes a fully integrated framework combining adaptive Riccati-based LQR with LSTM traffic forecasting. The traffic network is modeled as a discrete-time state-space system, where signal timings are dynamically updated using both real-time sensor feedback and predictive insights. Extensive simulations in the SUMO platform on a realistic multi-arm urban roundabout demonstrate notable improvements in waiting time, trip duration, and time loss compared to traditional Riccati and LQR methods. These results highlight the potential of hybrid model-based and learning-based control for scalable, adaptive traffic management in smart city applications.

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In summary, the key contributions of this work include:

- We propose a hybrid traffic signal control framework that integrates an adaptive Riccati-based LQR with a LSTM network for traffic state forecasting. This integration enables real-time, data-informed signal optimization across multiple intersections.
- The framework models urban traffic as a discrete-time state-space system, where the LQR controller dynamically adjusts gain matrices by solving the Riccati equation online, based on both current sensor feedback and short-term traffic predictions.
- We implement the system within a realistic simulation environment using SUMO, based on a detailed sketch of the Cong Truong Dan Chu roundabout and its surrounding network in Ho Chi Minh City. The design captures real-world traffic complexity and includes the use of virtual sensors and crowd-sourced traffic data.
- Through extensive simulation experiments, we demonstrate that the proposed method significantly outperforms traditional Riccati and Riccati+LQR controllers, achieving up to 29.39% reduction in trip duration, 57.97% reduction in average waiting time, and 42.64% reduction in time loss, while maintaining throughput.

The rest of the paper is organized as follows: The proposed approaches are presented in detail in Section II. Section III describes the experiments conducted. Finally, Section IV concludes the paper and discusses future work.

II. METHODOLOGY

Figure 1 illustrates the overall architecture of the proposed hybrid control framework, which integrates data-driven traffic forecasting with feedback-based signal control. The design reflects a closed-loop control system enriched with predictive capabilities to adapt more effectively to urban traffic dynamics.

At the top of the diagram, the LSTM model was pre-trained using historical traffic data collected from Waze [19], a crowd-sourced traffic platform. During simulation, it receives real-time traffic states from SUMO as input for inference, detailed in Subsection II-E, which captures temporal patterns and forecasts the short-term evolution of the traffic state, denoted by $\hat{x}$. The predicted state is then passed to the signal timing optimization module, described in Subsection II-C, where it supports proactive adjustments to green-split strategies.

Simultaneously, the SUMO environment functions as the control plant, emulating a multi-node traffic network, including dynamic interactions at the Cong Truong Dan Chu roundabout. Simulated sensor feedback, such as virtual loop detectors, provides real-time observations of the current traffic state. In a real-world deployment, this data would be sourced from physical infrastructure (e.g., loop detectors or cameras). Based on this feedback, an adaptive Riccati-based LQR controller, formulated across Subsections II-A, II-B, and II-D, computes state feedback gains by solving the Riccati equation online. These gains are used to regulate signal timings in conjunction with the LSTM-based forecasts, thereby integrating reactive and anticipatory control strategies.

The dual-stream architecture, combining predictive (open-loop) and reactive (closed-loop) control, ensures both anticipation of future traffic flows and robustness to unexpected disturbances. This hybrid approach is particularly advantageous in nonlinear, stochastic environments like urban intersections, where delays, surges, and irregular driver behaviors are common. By integrating forecasted traffic evolution with real-time sensor feedback, the system balances short-term responsiveness with long-term optimization, leading to more efficient and adaptive signal control.

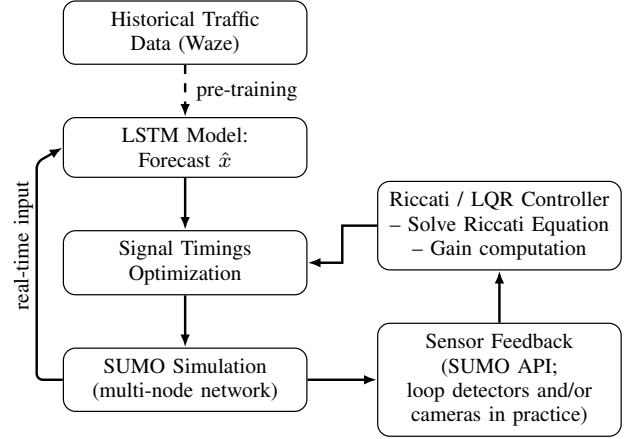


Fig. 1. Block diagram of the proposed hybrid control system. The LSTM model is pre-trained on historical data (Waze) and receives real-time traffic states from SUMO for forecasting. The forecast and feedback are used for signal optimization via LQR/Riccati control.

A. Traffic Network Dynamics Model (State-space model)

Assume the system has n intersections, each with m_i lanes controlled by traffic signals. Define the state vector as:

$$x_k = \begin{bmatrix} q_1(k) \\ q_2(k) \\ \vdots \\ q_N(k) \end{bmatrix} \in \mathbb{R}^N, \quad (1)$$

where $q_i(k)$ is the queue length (number of vehicles) at lane i at time k , and the total number of lanes is $N = \sum_i m_i$.

The control input vector is:

$$u_k = \begin{bmatrix} g_1(k) \\ g_2(k) \\ \vdots \\ g_N(k) \end{bmatrix} \in \mathbb{R}^N, \quad (2)$$

where $g_i(k)$ is the green light duration for lane i at time k .

The linearized state equation [20] describing the system dynamics is:

$$x_{k+1} = Ax_k + Bu_k + w_k, \quad (3)$$

where

- $A \in \mathbb{R}^{N \times N}$ is the state transition matrix, describing how current traffic influences the next state.

- $B \in \mathbb{R}^{N \times N}$ is the input matrix, representing the impact of green time on queue length changes.
- w_k is an unpredictable disturbance, modeled as Gaussian noise with $w_k \sim \mathcal{N}(0, W)$.

B. LQR Problem

We aim to find an optimal control sequence $\{u_k\}$ minimizing the infinite-horizon cost function:

$$J = \sum_{k=0}^{\infty} (x_k^\top Q x_k + u_k^\top R u_k), \quad (4)$$

where

- $Q \in \mathbb{R}^{N \times N}$ is the state weighting matrix, positive semidefinite, emphasizing queue length reduction (congestion mitigation).
- $R \in \mathbb{R}^{N \times N}$ is the control weighting matrix, positive definite, limiting rapid changes in signal timing to ensure stability.

C. Optimal Control Solution

The optimal linear state feedback control law is given by:

$$u_k = -K x_k, \quad (5)$$

where K is calculated from the matrix P , which is the solution of the discrete-time Algebraic Riccati Equation (ARE) [21], [22]:

$$P = A^\top P A - A^\top P B (R + B^\top P B)^{-1} B^\top P A + Q. \quad (6)$$

Once P is obtained, the feedback gain matrix K is computed by:

$$K = (R + B^\top P B)^{-1} B^\top P A. \quad (7)$$

D. Adaptive Riccati Solution

For time-varying traffic environments, we apply an *Adaptive LQR* approach that updates A, B, Q, R continuously based on real data:

- Collect measurements of x_k, u_k via sensors.
- Use Recursive Least Squares (RLS) algorithm to estimate system matrices A_k, B_k :

$$\hat{\theta}_k = \hat{\theta}_{k-1} + L_k (y_k - \phi_k^\top \hat{\theta}_{k-1}), \quad (8)$$

where $\hat{\theta}$ stacks the unknown parameters of A and B , y_k is the observed next state, ϕ_k is the regression vector containing past states and controls, and L_k is the adaptive gain.

- Using updated $\hat{A}_k, \hat{B}_k$, solve the Riccati equation (6) online to update P_k and K_k in real-time.

E. Traffic Flow Forecasting: Dataset and Modeling Approach

The dataset consists of structured JSON responses from the Waze API, each representing a timestamped route query. Each query contains an “alternatives” field listing potential routes with metadata such as route name, unique IDs, and key metrics including estimated travel times under real-time, historical, and free-flow conditions. Detailed segment-level information is provided in the “results” array, including geographic coordinates (“path”), segment IDs, road classification codes, segment length (meters), and road names. Raw timestamps are decomposed into structured temporal features (e.g., time-of-day, day-of-week, week ID) to capture daily and seasonal traffic patterns. The JSON data is parsed and converted into tabular format (e.g., CSV) with columns such as date_ID, DIST, DIST_gap, LATS, LNGS, TIME, time_gap, time_ID, and WEEK_ID, while multi-valued fields like coordinate sequences are retained as lists within cells to preserve spatial-temporal resolution.

Traffic data were collected at five-minute intervals, yielding twelve samples per hour and approximately 204 queries per origin-destination pair over a 17-hour daily period. Over approximately 1.5 months (≈ 45 days), the system amassed roughly 226,800 trajectory records, each corresponding to a single trip in JSON format. This comprehensive dataset serves as the foundation for training deep neural networks, specifically LSTM models [23]. Each input sample is a multivariate feature vector including traffic state variables such as vehicle flow counts per lane or segment, average speeds, and traffic density or occupancy rates, combined with temporal features to capture typical traffic patterns. Additional contextual data, such as weather conditions and traffic incidents, are incorporated when available to improve prediction accuracy. The dataset is split into 70% for training, 15% for validation, and 15% for testing, ensuring robust model development and evaluation. Before training, normalization and missing data handling ensure numerical stability and robustness.

The model forecasts short-term traffic states 15 to 30 minutes ahead using historical data from the previous 30 to 60 minutes. Inputs consist of consecutive state vectors from time steps $k-n$ to k , where n denotes the look-back window length, as described in equation (9). The output consists of predicted traffic states at future time steps $\hat{x}_{k+1|k}, \hat{x}_{k+2|k}, \dots$, which are used to update and adapt the traffic signal controller parameters. Formally, the one-step ahead prediction can be expressed as

$$\hat{x}_{k+1|k} = \text{LSTM}(x_{k-n}, x_{k-n+1}, \dots, x_k) \quad (9)$$

where n denotes the length of the look-back window of past traffic states used as input. This dataset structure enables the LSTM to capture nonlinear temporal dependencies and complex traffic dynamics, yielding accurate forecasts that support adaptive, timely traffic signal control. Such forecasting capability enhances the accuracy of state prediction and enables more effective parameter estimation, which in turn facilitates the adaptive updating of the weighting matrices Q and R in equation (4). Consequently, the traffic signal controller can be dynamically reconfigured to optimize signal

timing based on the predicted traffic conditions, leading to enhanced traffic flow management.

III. EXPERIMENTS

A. Experiment Objectives

The experiment is designed to evaluate the effectiveness of a hybrid traffic signal control strategy that combines an adaptive Riccati-based LQR with traffic state forecasting via a pre-trained LSTM network. The integrated system operates in a closed-loop configuration, using real-time sensor feedback from the simulation environment and short-term traffic predictions to optimize signal timings dynamically across multiple intersections. To assess the benefits of this hybrid approach, its performance is systematically compared against two baselines: a traditional Riccati-only controller and a Riccati + static LQR configuration, both implemented under identical simulation conditions.

B. Experimental Methodology and Simulation Environment

This study employs SUMO [24], an open-source traffic simulation platform that supports multi-node network models and customizable controllers via its Application Programming Interface (API). Vehicle flows were generated using probabilistic demand models, with arrival rates varying over time to mimic peak and off-peak traffic conditions. To test the system's adaptability, asymmetric traffic volumes were assigned to different directions, creating imbalanced inflow scenarios. Real-time vehicle positions and speeds were collected at 1-second intervals via virtual loop detectors to feed the prediction and control modules.

The simulation domain is based on a realistic model of the Cong Truong Dan Chu area in Ho Chi Minh City, Vietnam, featuring a prominent multi-arm roundabout located at the junction of Vo Thi Sau, Cach Mang Thang Tam, Ba Thang Hai, Ly Chinh Thang, Nguyen Phuc Nguyen, and Nguyen Thuong Hien streets. The surrounding area contains multiple intersections, most of which are controlled by traffic signals, consistent with real-world conditions. Covering an area of more than 5 km² with an approximate radius of 1.5 km centered on the roundabout, this major urban node connects central Districts 3 and 10. It experiences high traffic density, especially during peak hours, with frequent congestion and mixed vehicle flows including motorbikes, cars, and buses. Planned infrastructure developments, such as proposed steel overpasses, underground tunnels, and Metro Line 2 intended to run beneath the roundabout, currently in the stages of planning and site clearance, further highlight the complexity and long-term significance of traffic conditions in this area. The simulation reflects this complexity by modeling heterogeneous vehicle types, varied inflow rates, and signalized intersections at each arm of the roundabout. Such a realistic setting enables a rigorous evaluation of control methods under conditions representative of urban traffic at one of Ho Chi Minh City's most critical and congested junctions, as illustrated in Figure 2.

However, while traffic signals are present at the central roundabout, in reality it is often managed by traffic police during peak hours, operating primarily under priority and



Fig. 2. Simulation sketch of the Cong Truong Dan Chu area in Ho Chi Minh City. The modeled region covers more than 5 km², centered at the roundabout and extending within a 1.5 km radius. The area includes multiple major arterial roads and signalized intersections, representing a realistic and high-density urban traffic environment.

yield rules rather than signal control. To enable a systematic evaluation of signal-based control methods. Within the simulation environment, traffic signals were artificially installed exclusively on the two lanes approaching the roundabout from Ba Thang Hai Street, as illustrated in Figure 3. This addition allows for the implementation and assessment of various control strategies.

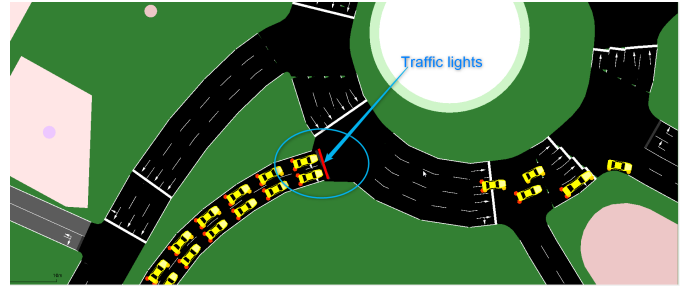


Fig. 3. Location of artificially added traffic signals on the two lanes from Ba Thang Hai Street into the roundabout.

Maintaining existing traffic signals in the surrounding area ensures the simulation preserves realistic urban traffic dynamics, while the signalization of the roundabout creates a controlled framework to rigorously compare control algorithms. This design choice facilitates the analysis of traffic metrics such as average speed, waiting time, and throughput under consistent and comparable conditions.

Although the signalization of the roundabout is a hypothetical modification, it reflects a plausible scenario for future traffic management interventions or temporary control measures. Thus, this experimental setup balances fidelity to real-world traffic patterns with the practical requirements for evaluating advanced traffic signal control strategies.

C. Evaluation Metrics

The evaluation of control methods is conducted using several key performance metrics that comprehensively reflect the

TABLE I
COMPARISON OF TRAFFIC CONTROL METHODS WITH IMPROVEMENTS OVER RICCATI BASELINE

Metric	Riccati	Riccati + LQR	LSTM	Improvement [†] vs. Riccati (%)
Avg. Speed (m/s)	9.87	9.89	10.65	+0.20 / +7.90
Avg. Trip Duration (s)	174.49	152.32	123.21	-12.71 / -29.39
Avg. Waiting Time (s)	81.78	60.15	34.37	-26.45 / -57.97
Avg. Time Loss (s)	119.91	97.75	68.78	-18.48 / -42.64
Avg. Departure Delay (s)	6.94	6.94	6.96	0.00 / +0.29
Vehicle Trips Completed	409	409	409	0.00 / 0.00
Vehicles Loaded	431	431	431	0.00 / 0.00
Total Route Length (m)	1123.96	1122.45	1124.24	-0.13 / +0.02
Total Travel Time (s)	71365.00	62301.00	50394.00	-12.70 / -29.39*
Total Departure Delay (s)	2837.29	2839.13	2848.29	+0.06 / +0.39

Note[†]: Improvement is shown as Riccati+LQR / LSTM compared to Riccati. Negative values indicate a decrease (improvement in delay/duration); positive values indicate an increase (e.g., speed). Improvement percentages are computed as: Improvement (%) = $\frac{\text{New Value} - \text{Riccati Value}}{\text{Riccati Value}} \times 100$

*Total travel time may decrease with shorter durations, even with the same number of trips.

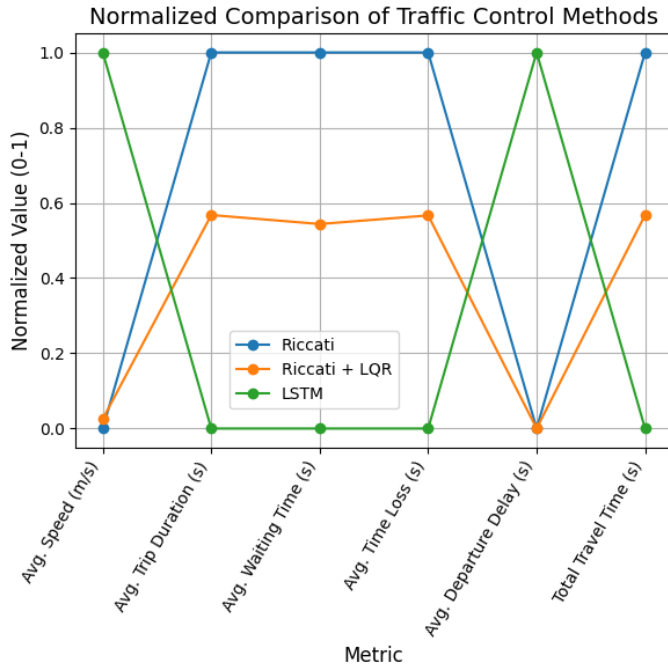


Fig. 4. Normalized comparison of traffic control methods across key performance metrics. Each metric is scaled to a [0–1] range to highlight relative performance differences between Riccati, Riccati + LQR, and LSTM-based control. Higher normalized values indicate better performance for metrics such as speed, and worse performance for metrics like delay or time loss.

effectiveness of traffic management strategies. These metrics include average speed, average trip duration, average waiting time, number of vehicle trips completed, and total time loss. Each metric provides insight into different aspects of system performance, such as traffic flow efficiency, delay reduction, and throughput capacity. Table I summarizes the comparative results of these metrics for the Riccati, Riccati + LQR, and LSTM-based control methods, highlighting their respective improvements and limitations.

D. Comparative Analysis of Control Method Effectiveness

To better understand the performance of each traffic control strategy, key operational metrics are analyzed in this subsection based on the data presented in Table I. Additionally,

Figure 4 provides a normalized overview of the results, highlighting the relative effectiveness of the Riccati, Riccati + LQR, and LSTM-based controllers.

Average Speed: The Riccati controller achieves 9.87 m/s, slightly increasing to 9.89 m/s (+0.20%) with LQR feedback. The LSTM controller further boosts speed to 10.65 m/s (+7.90%), indicating smoother traffic flow, likely due to its capacity to model temporal dynamics and adapt to varying traffic patterns.

Average Trip Duration: Average trip time drops from 174.49 s (Riccati) to 152.32 s with LQR (–12.71%), and further to 123.21 s (–29.39%) using LSTM. The reduction reflects more efficient vehicle movement and fewer stops, especially under the adaptive LSTM policy.

Average Waiting Time: Waiting time improves from 81.78 s (Riccati) to 60.15 s with LQR (–26.45%), and drastically to 34.37 s (–57.97%) with LSTM, highlighting its superior congestion prediction and signal timing.

Time Loss: Time loss reduces from 119.91 s (Riccati) to 97.75 s (–18.48%) with LQR, and to 68.78 s (–42.64%) with LSTM. This reflects LSTM’s efficiency in minimizing delays caused by stops and deceleration through learned control behavior.

Vehicle Trips Completed: All three methods, Riccati, Riccati + LQR, and LSTM, complete 409 vehicle trips within the 3000-second simulation. The uniform time frame here ensures fair comparison, highlighting LSTM’s efficiency gains in speed, delay, and duration rather than throughput alone.

Despite achieving the best overall performance in speed, waiting time, and trip duration, the LSTM-based control exhibits a slightly higher average departure delay (6.96 s) compared to the Riccati-based methods (6.94 s), likely due to its proactive signal timing adjustments that may prioritize flow optimization over minimizing initial vehicle departure latency.

LSTM-based control consistently outperforms traditional methods across most key metrics, particularly in speed, trip duration, waiting time, and time loss. While Riccati + LQR offers noticeable improvements over the baseline, LSTM’s temporal modeling capabilities allow for better adaptation to dynamic traffic conditions and more effective delay reduction.

These results underscore the advantages of integrating classical control with deep learning. In intelligent transportation

systems, combining structured feedback (e.g., LQR) with sequence-based models like LSTM enables real-time, adaptive, and robust traffic control under nonlinear, evolving scenarios.

IV. CONCLUSION

This study presents a comprehensive evaluation of traffic signal control strategies in a multi-intersection urban setting, comparing classical approaches (Riccati, Riccati + LQR) with an LSTM-enhanced control scheme that integrates deep learning-based traffic prediction. Microsimulation results show that:

- Riccati + LQR improves upon the baseline by incorporating state feedback, yielding better responsiveness and reduced delays. The LSTM-enhanced controller, combining predictive learning with adaptive Riccati-based control, outperforms both classical methods across nearly all key metrics (average speed, trip duration, waiting time, and time loss).
- By introducing temporal foresight, the LSTM model enables anticipatory signal control in response to predicted traffic flows. Identical throughput across all methods confirms that performance gains result from control efficiency rather than differing traffic demand.

These findings highlight the potential of hybrid control architectures that combine the structural rigor of model-based techniques like LQR with the adaptability of deep learning. Such integration is well-suited for intelligent transportation systems, offering scalable, data-driven solutions to complex and nonlinear traffic dynamics.

Future research may explore several promising directions, including extensions to reinforcement learning, multimodal models that integrate public transport and pedestrian flows, and deployment strategies that combine online adaptation with offline training to enhance real-world robustness. Additionally, improving system resilience to unexpected disruptions (e.g., accidents) remains critical, potentially through fallback mechanisms such as anomaly detection. Given the computational demands of Riccati and LSTM components, future work should also consider latency profiling and lightweight deployment strategies. More realistic evaluation settings, incorporating weather variability, sensor noise, and traffic disturbances, are necessary to validate system performance. Exploring the transferability of the proposed approach to larger, heterogeneous traffic networks would help assess its scalability. Finally, empirical comparisons with MPC and deep reinforcement learning methods would provide a stronger benchmark beyond Riccati-based baselines.

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